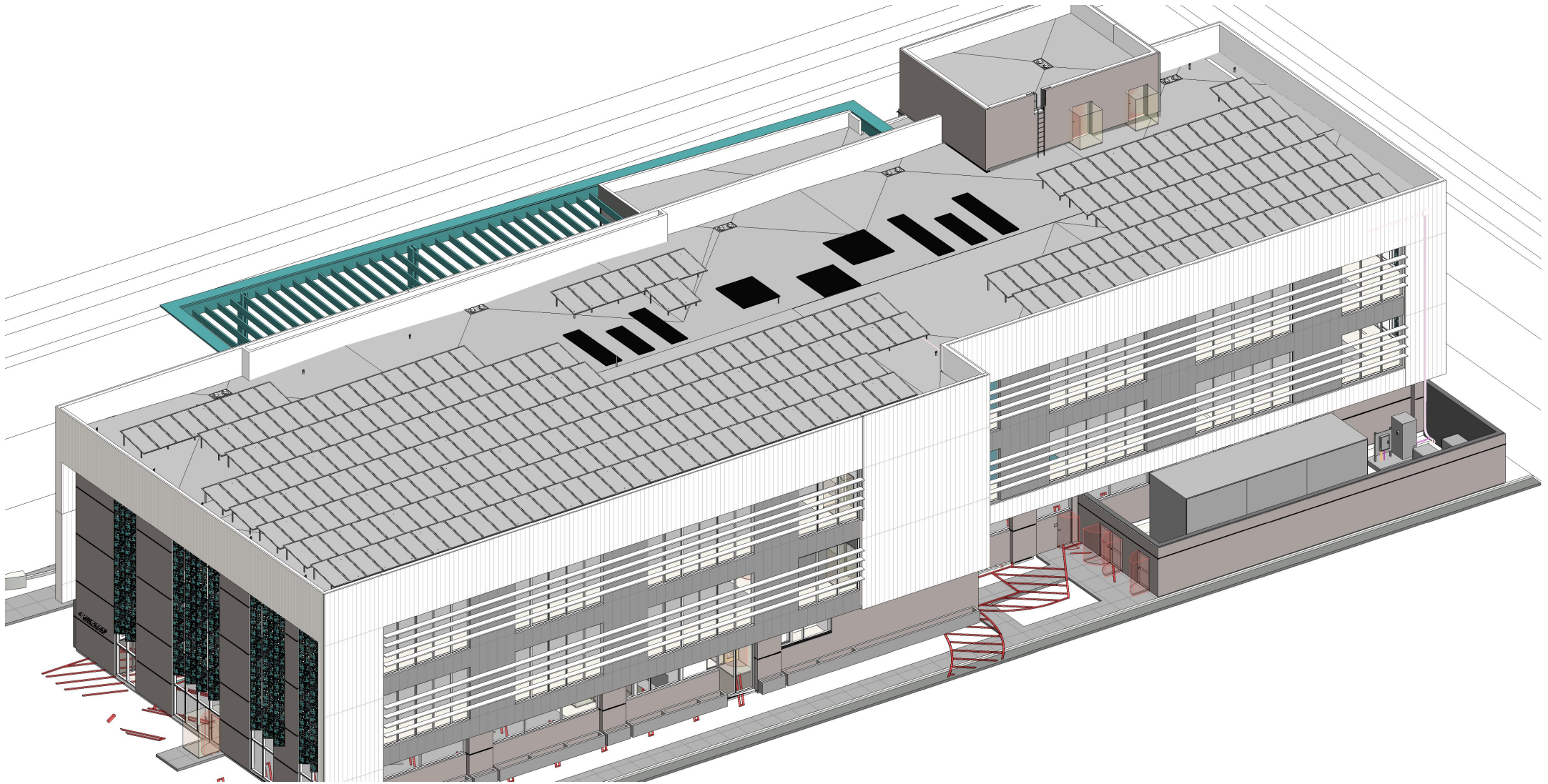
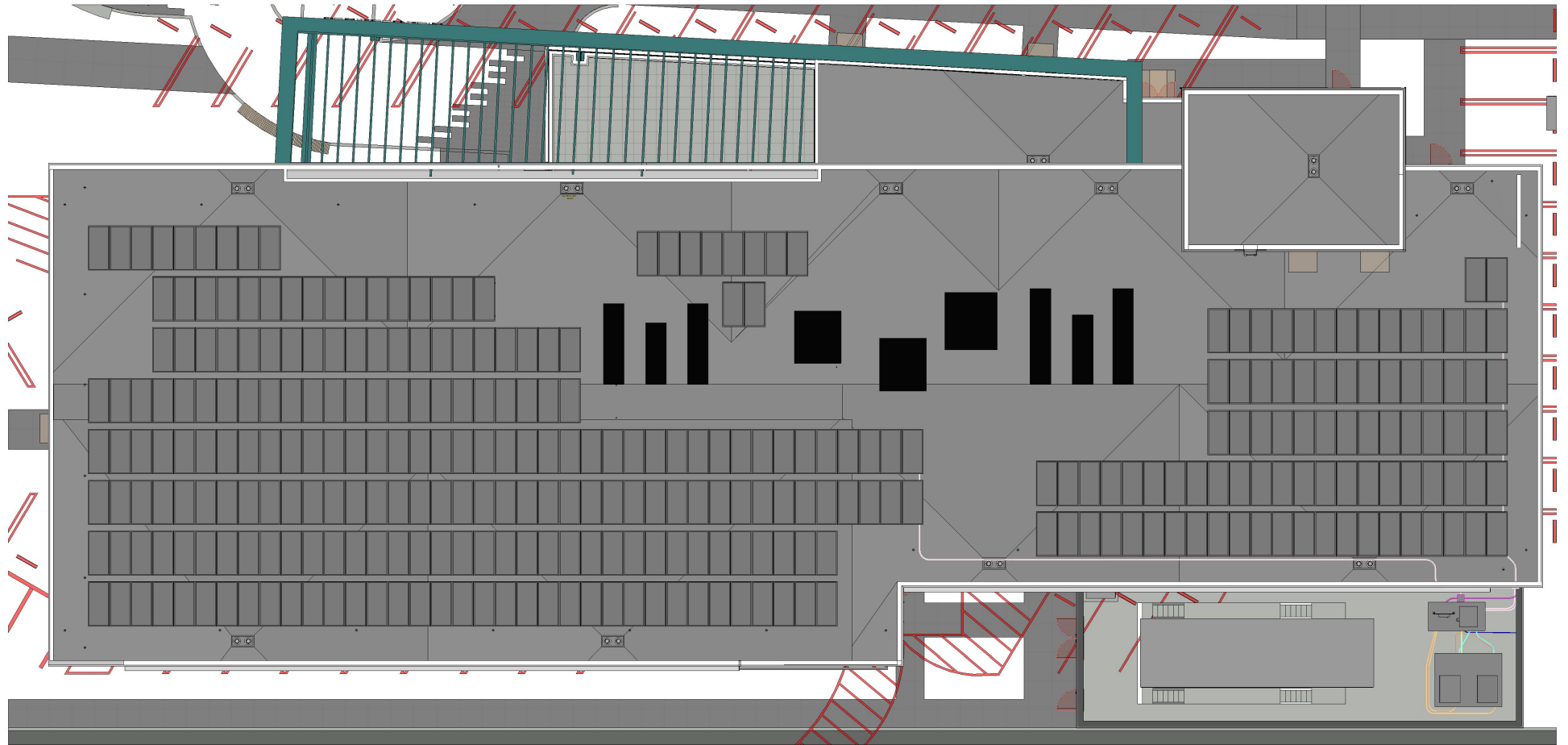




Solar BIM Coordination Model



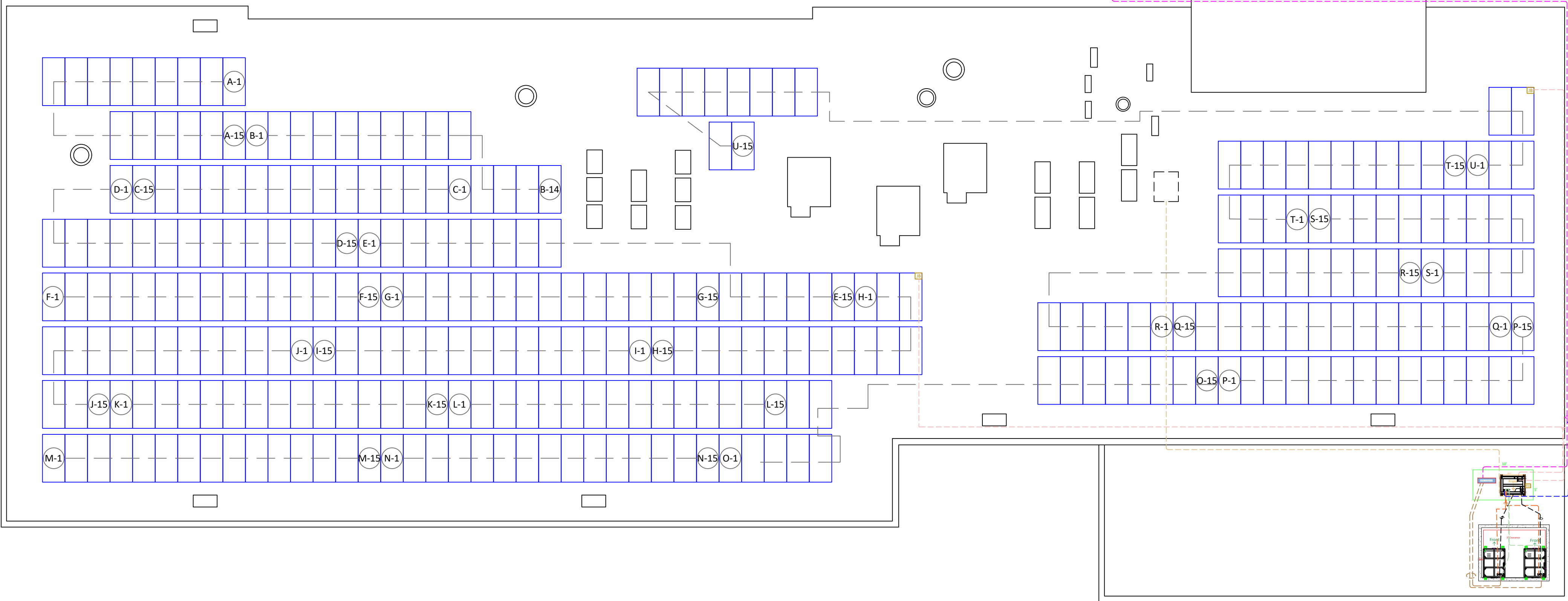
Top view of Electrical BIM Model Submission

BILL OF MATERIALS		
EQUIPMENT	QTY	DESCRIPTION
SOLAR PV MODULE	314	HANWHA Q.PEAK DUO XL-G11.3/BFG (590W)
INVERTER	1	RENON MPPT PV (R-MC600PVC01-US) (240KW / 6 POWER MODULES PV CAPACITY)
OPTIMIZERS	314	TIGO TS4-X-F (1000V) OPTIMIZERS
BATTERY	2	RENON POWER MPACK 233A (R-MP233125A1-US)
COMBINER PV	1	RENON MCOMBINER PV (R-MC600PVC01-US) 240KW
RAPID SHUTDOWN SWITCH	1	RAPID SHUTDOWN SWITCH
JUNCTION BOX	2	JUNCTION BOX, 1000V, NEMA 3R, UL LISTED
SOLAR LOAD CENTER	1	SOLAR LOAD CENTER 600A RATED, 480V, 3-PHASE, 4-W, NEMA 3R, UL LISTED
ATTACHMENT	1184	IRONRIDGE TILT LEG SYSTEM & U-FOOT
ATTACHMENT	1184	BOLT, BONDING 3/8-16 SQ HEAD, 2.25"
ATTACHMENT	1184	NORTH TILT LEG, 1.5" SQ, LENGTH VARIES
ATTACHMENT	3552	NUT, FLANGE HEX 3/8-16 SS
ATTACHMENT	1184	BOLT, BONDING 3/8-16 SQ HEAD, 1.0"
ATTACHMENT	1184	PRE-ASSEMBLED SOUTH TILT LEG
ATTACHMENT	1184	WASHER, FLAT 3/8 SS
ATTACHMENT	1184	BOLT, 3/8-16 X 2.5" CS SST
ATTACHMENT	1184	U-FOOT
SUPPORT STRUCTURE	1184	HSS 6X6X3/8 PER STRUCTURAL PLANS
SUPPORT STRUCTURE	1184	3-1/2" PIPE PER STRUCTURAL PLANS
RAIL	167	IRONRIDGE XR-100 RAIL (168")
BONDED SPLICE	134	BONDED SPLICE
CLAMPS	612	UNIVERSAL FASTENING OBJECTS (UFO)
CLAMPS	64	STOPPER SLEEVE
GROUNDING LUG	16	GROUNDING LUG

ARRAY WEIGHT (LOAD CALC'S)		
Number of Modules	314	
Module Weight	67.68	LBS
Total Module (Array) Weight	21251.52	LBS
Number of Attachment point	1184	
Mounting System Weight (Per Module)	1	LBS
Mounting System Weight	1184.00	LBS
Total System Weight (Module Weight + Mounting System Weight)	22435.52	LBS
Weight at Each Attachment Point (Array Weight / Number of Attachment Point)	17.95	LBS
Module Area (95.11"x44.64")	29.48	SqFt
Total Array Area	9258.01	SqFt
Distributed Load (Total System Weight / Total Array Area)	2.42	Per SqFt
Total Roof Area	24946	SqFt
Total Percentage of Roof Covered (Total Array Area / Total Roof Area) x 100	37.11%	

NOTE: TWO (2) ADDITIONAL MODULES WILL BE PROVIDED FOR MAINTENANCE PURPOSES.

N CHERRY AVE.
BACK OF BUILDING

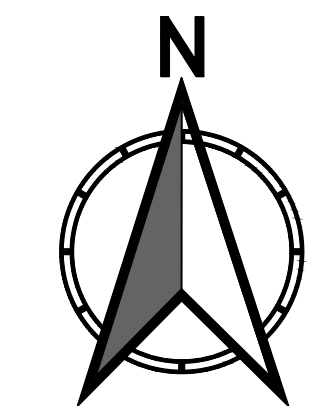


LEGEND	
MSP	MAIN SERVICE PANEL
MCP	RENON MCOMBINER PV
BAT	RENON POWER MPACK 233A
SLC	SOLAR LOAD CENTER
RSS	RAPID SHUTDOWN SWITCH
JB	JUNCTION BOX
	VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
	EMT CONDUIT

STRING LEGEND					
A	STRING #1- 15 MODULES	J	STRING #10- 15 MODULES	S	STRING #19- 15 MODULES
B	STRING #2- 14 MODULES	K	STRING #11- 15 MODULES	T	STRING #20- 15 MODULES
C	STRING #3- 15 MODULES	L	STRING #12- 15 MODULES	U	STRING #21- 15 MODULES
D	STRING #4- 15 MODULES	M	STRING #13- 15 MODULES		
E	STRING #5- 15 MODULES	N	STRING #14- 15 MODULES		
F	STRING #6- 15 MODULES	O	STRING #15- 15 MODULES		
G	STRING #7- 15 MODULES	P	STRING #16- 15 MODULES		
H	STRING #8- 15 MODULES	Q	STRING #17- 15 MODULES		
I	STRING #9- 15 MODULES	R	STRING #18- 15 MODULES		

CONDUIT LEGEND	
	JUNCTION BOX TO MCP
	MCP TO BATTERY #1
	MCP TO BATTERY #2
	SLC TO ACD
	IDF TO MCP
	EMT AUX TO MCP
	AUX POWER CONDUIT
	BATTERY TO SLC
	COMMUNICATION CONDUIT

FRONT OF BUILDING
N SULTANA AVE.

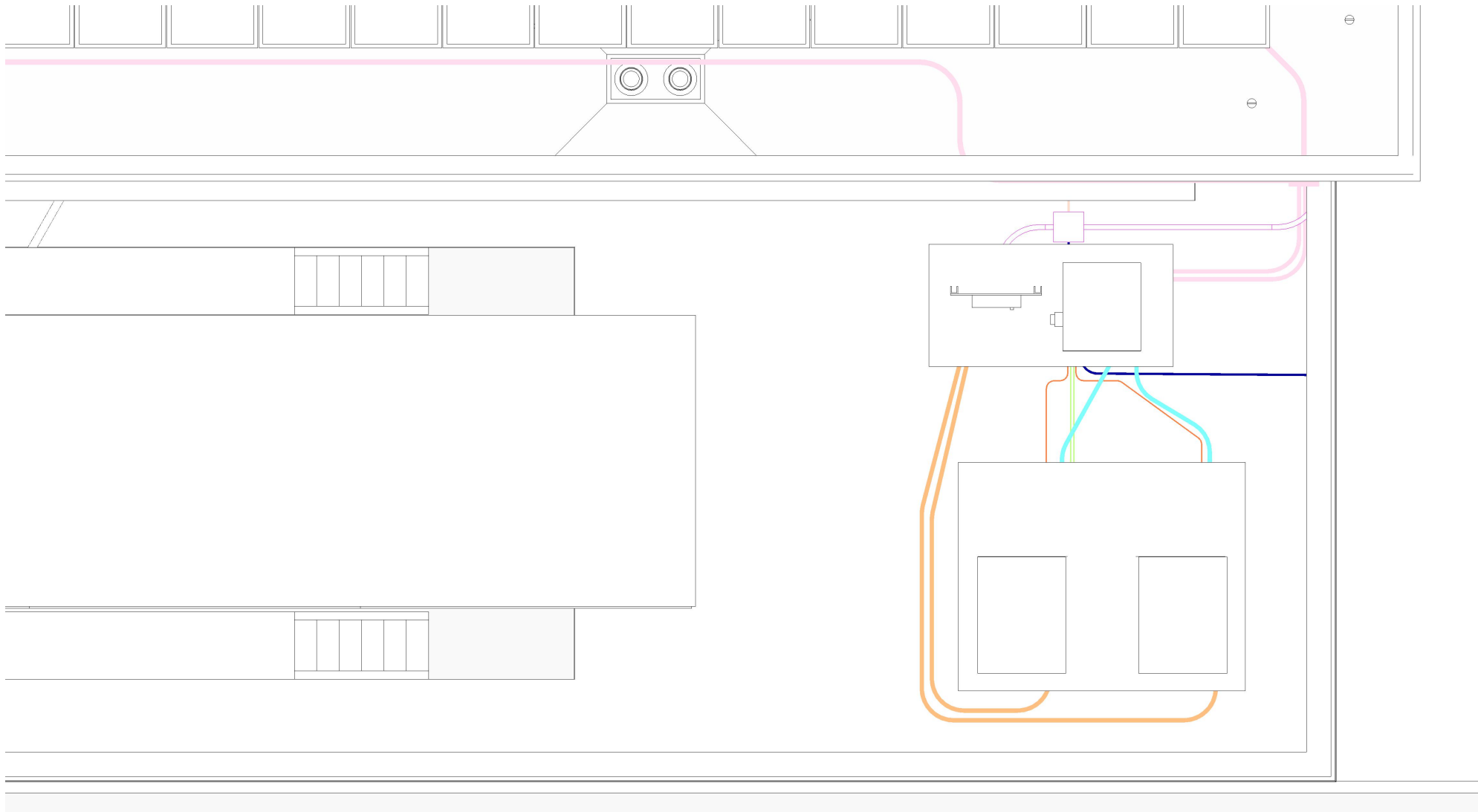


1/12"=1'-0"

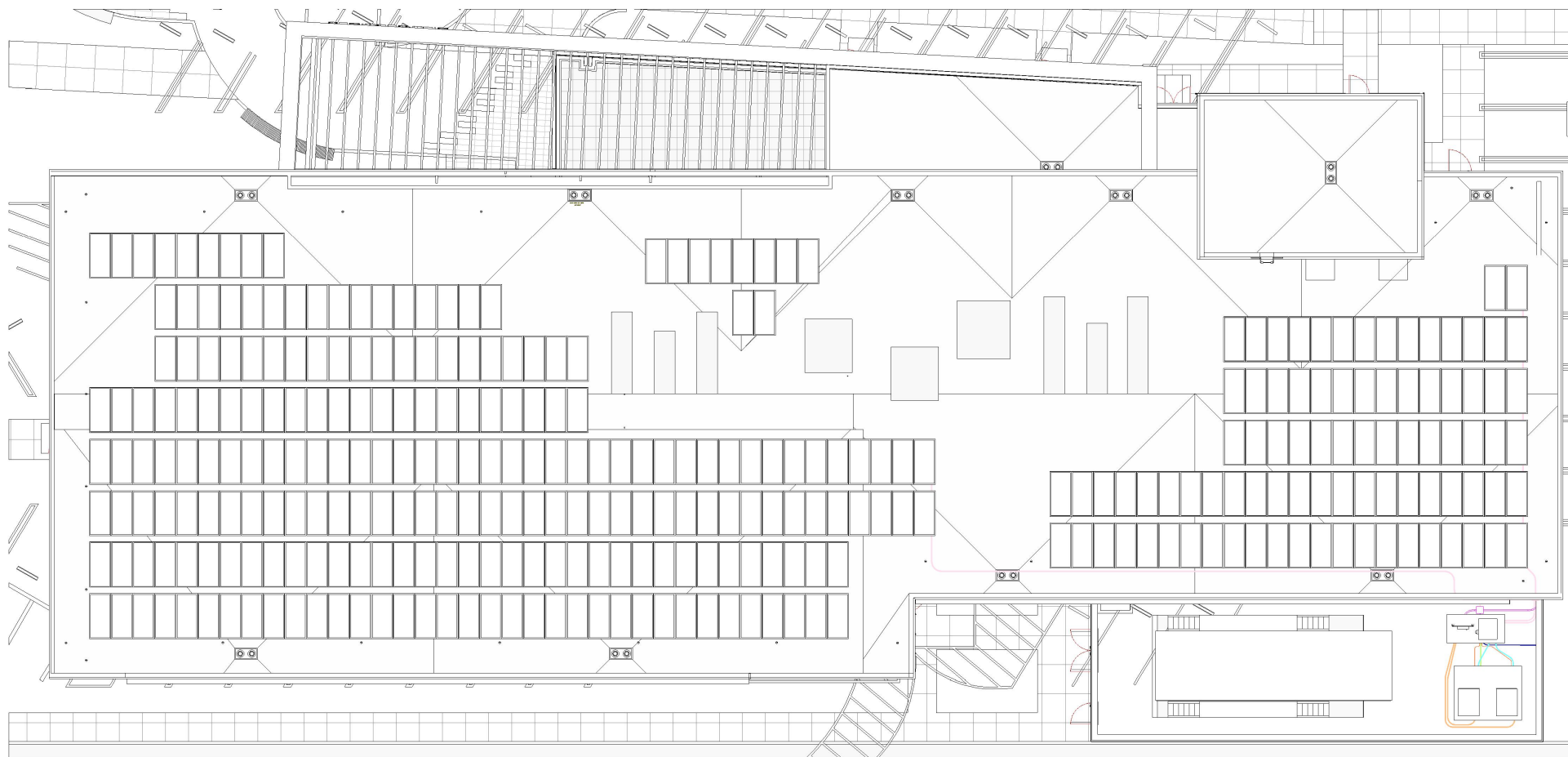
311 N SULTANA AVE COMMERCIAL 311 N SULTANA AVE. ONTARIO, CA 91764

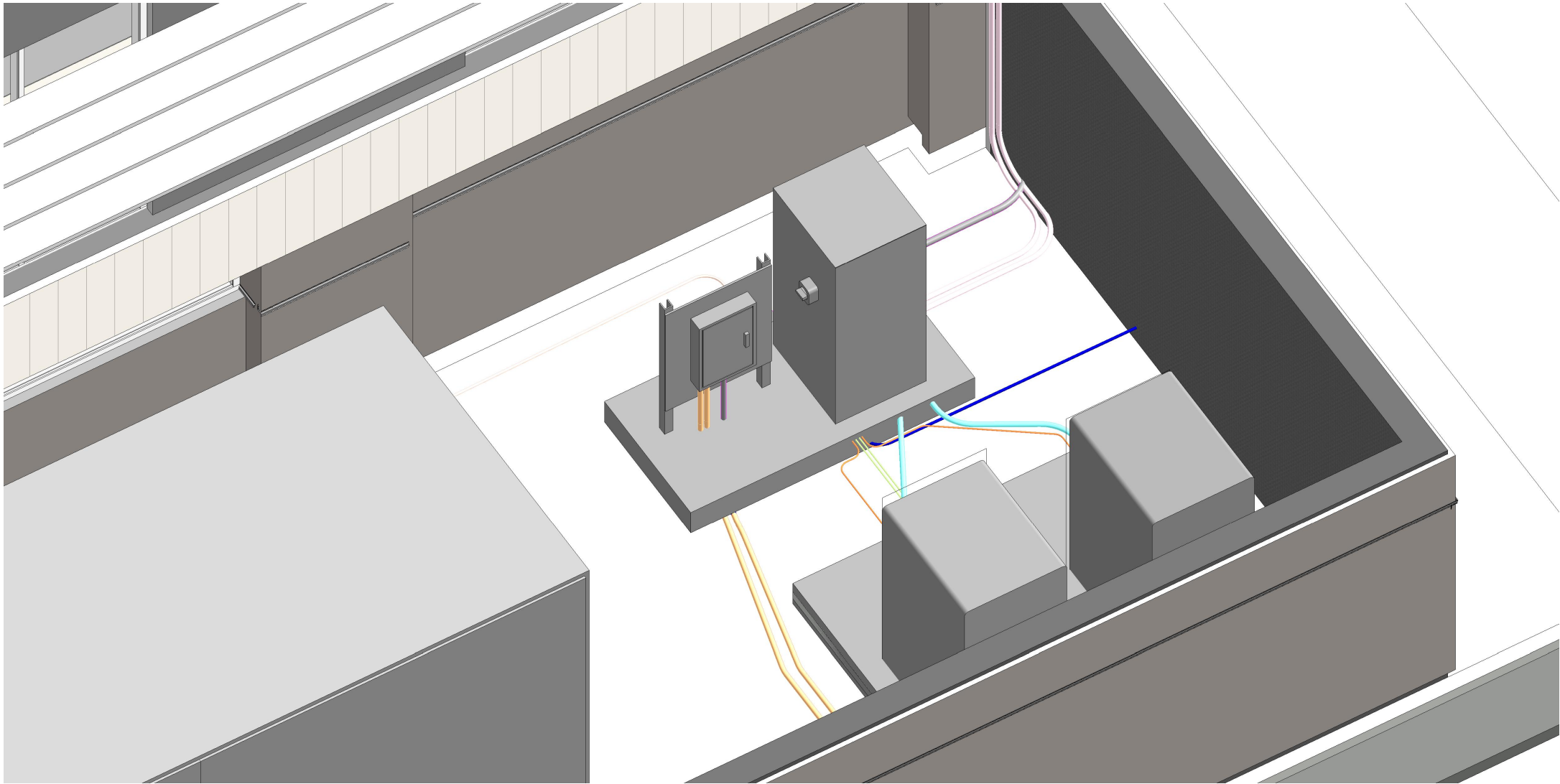
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PV-3
SHEET NAME
BOM & STRING LAYOUT
SHEET INFORMATION

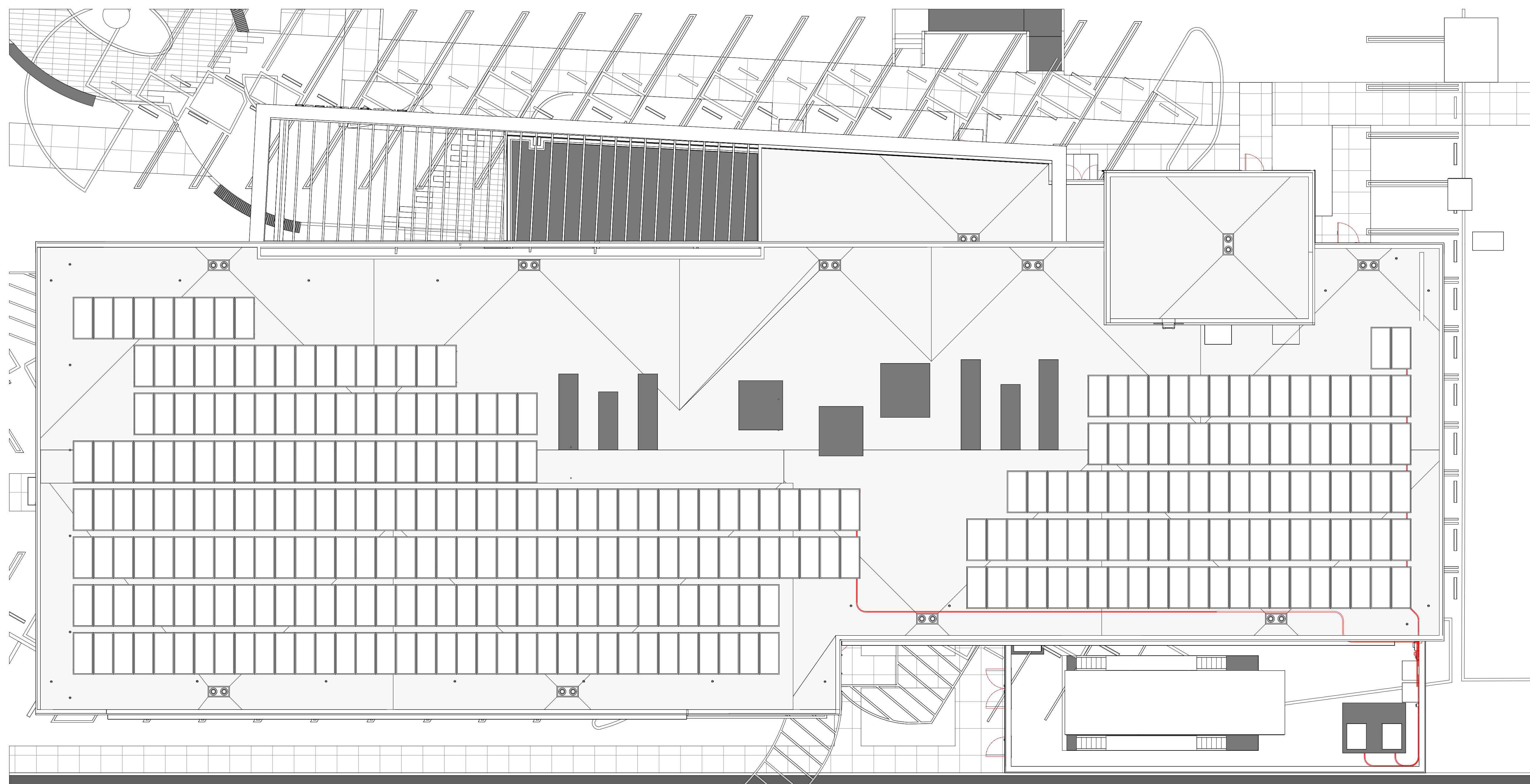


PV, BESS & Conduit Layout BIM Package

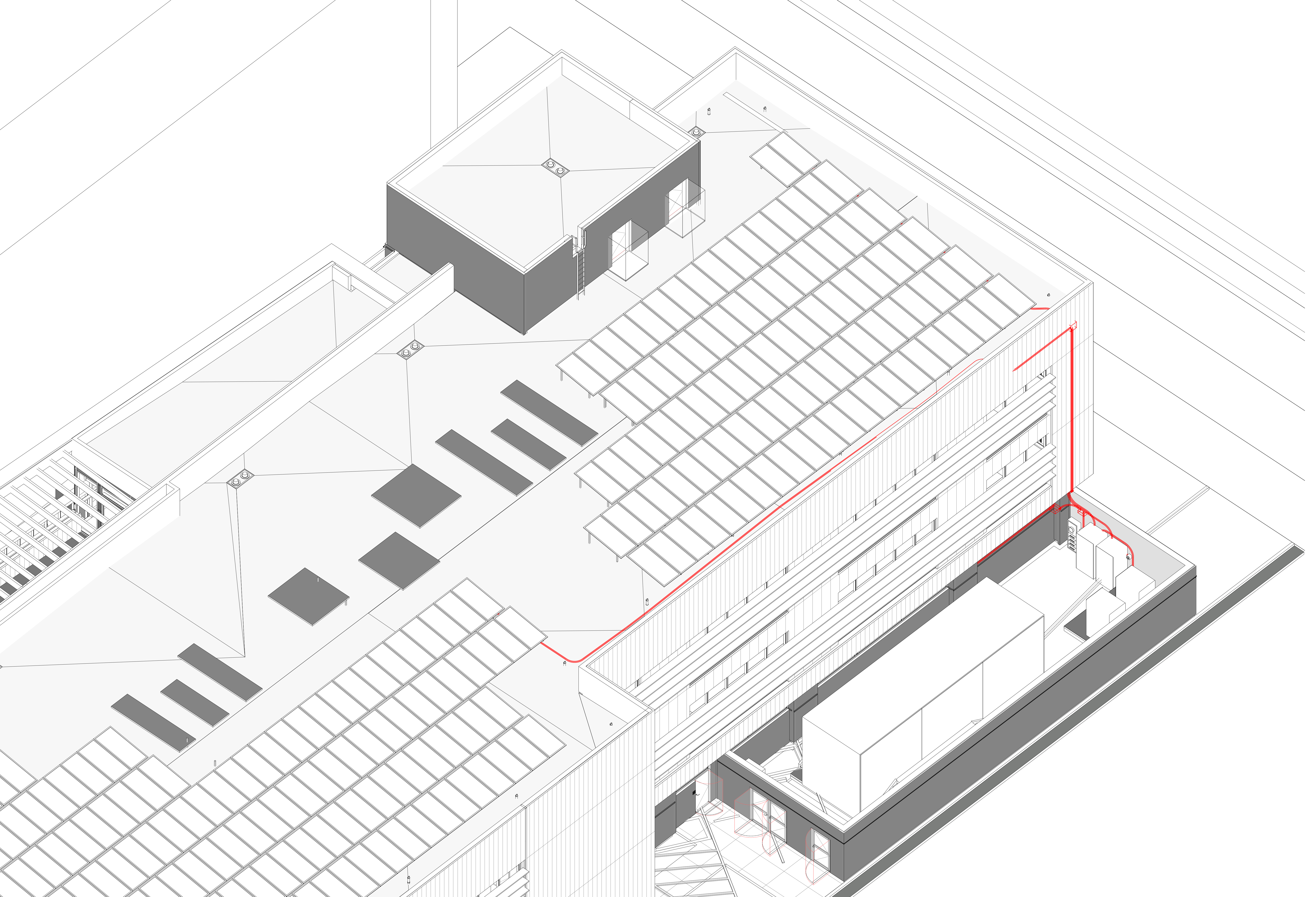


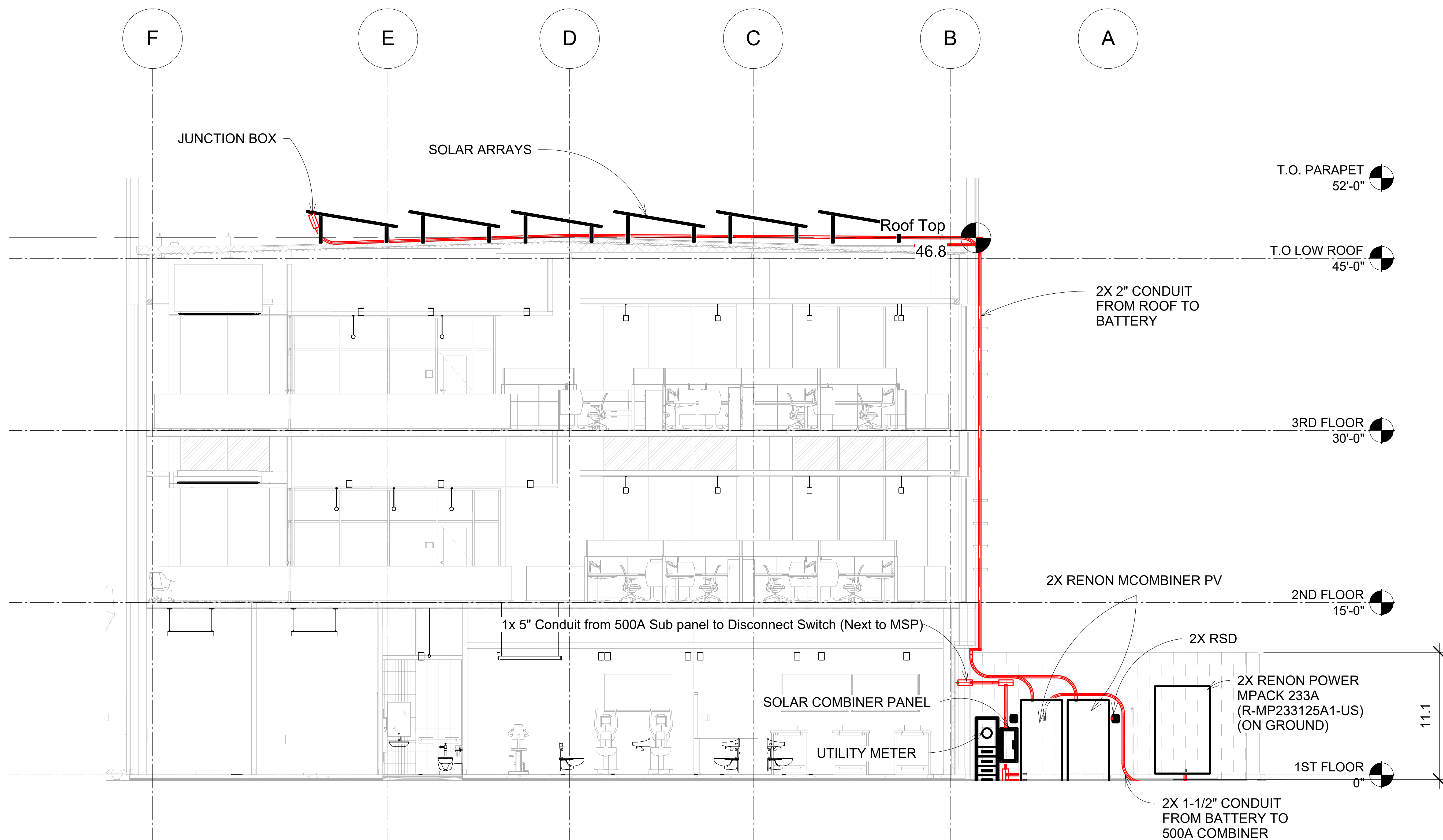


Equipment Placement & Conduit Coordination



Solar Array PV conduit route





PV EQUIPMENT WEST SIDE CROSS SECTION VIEW

